AI-Powered Unified Sri Lankan Tourism Platform Using GraphRAG and LLM Orchestration

Project Proposal Report R26-IT-050

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Abstract

Sri Lanka's tourism industry is rapidly recovering, but independent travelers still face significant challenges when planning personalized, time efficient travel routes due to the overwhelming number of attractions and static map limitations. Current digital platforms offer generic itineraries that fail to account for individual user preferences, spatial constraints, and realistic travel times, often leading to suboptimal and exhausting travel experiences. To address these critical limitations, this research proposes a Context Aware Spatiotemporal Itinerary Optimization System as a core component of a unified Sri Lankan tourism platform.

By leveraging custom developed Machine Learning (ML) profiling and metaheuristic routing models, the proposed system autonomously filters Points of Interest (POIs) based on tourist preferences and dynamically calculates a reachable geospatial travel radius. Developed using the MERN stack (MongoDB, Express.js, React.js, Node.js) and a dedicated Python microservice, the core research contribution is a custom Genetic Algorithm that processes spatial constraints to find the shortest, most efficient travel paths. Finally, the system integrates the Gemini API strictly as an Explainable AI (XAI) text formatting layer. This layer simply translates the complex mathematical map coordinates generated by the Genetic Algorithm into clear, human readable, day by day travel plans. Ultimately, this system aims to replace traditional manual trip planning with a personalized, dynamically optimized travel ecosystem, ensuring a highly efficient, enjoyable, and seamless touring experience in Sri Lanka.

Keywords: Spatiotemporal Itinerary Optimization, Machine Learning Profiling, Genetic Algorithms, Explainable AI (XAI), Dynamic Radius Generation, MERN Stack.

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1. Introduction

Sri Lanka’s tourism industry is experiencing rapid growth, with many international travelers choosing to explore the country independently. However, a major challenge these tourists face is planning their daily trips from their booked hotels. Travelers often have limited time, like 3 or 4 days, and they do not know how far they can practically travel from their current hotel location. They have to depend on multiple, disconnected sources to figure out if a place is reachable in one day, which complicates travel planning and diminishes the overall travel experience.

A critical flaw in the current tourism ecosystem is the heavy reliance on generic map applications and static recommendation websites. These conventional systems only show the path from point A to point B or just show a list of famous places. They do not reflect the specific constraints of the tourist, like the maximum distance (radius) they can travel in one day and return to the hotel. Furthermore, current platforms lack the intelligent reasoning required to match the tourist's personal preferences (like beaches or historical sites) with the travel time to provide a proper day by day travel plan.

To address these critical limitations, this research proposes a Context Aware Spatiotemporal Itinerary Optimization System for the Sri Lankan tourism sector. This system aims to replace unreliable static map searching with a dynamic, optimized travel routing ecosystem. By leveraging Machine Learning and geospatial analysis, the system automatically reads the user's interests, their hotel location, and their available time. It autonomously interprets this data to identify the best places to visit within a reachable radius and creates an optimized travel route, without requiring a human travel agent.

Developed utilizing modern web technologies including React.js, Node.js, Express, and MongoDB, along with Python, the system processes this user data and location coordinates. It uses machine learning for preference profiling and shortest path algorithms (like Genetic Algorithms) to calculate the most time saving route. Based on these calculations, the platform provides clear, proactive itineraries to tourists—such as suggesting the best connected places to visit without wasting time on the road—thereby ensuring a stress free and reliable travel experience in Sri Lanka.

1.1 Background and Literature Survey

Modern digital travel guidance is very different from traditional travel agencies because travelers today rely on internet maps and apps to make their travel decisions. In Sri Lanka, independent tourism is growing fast, but tourists face big challenges because they have to rely on generic map applications like Google Maps. Existing systems largely fail to address the real practical problems of a tourist, like how far they can travel from their booked hotel in a single day and still cover their favorite places. We need a reliable way to capture the user's interests and calculate an optimized daily travel plan before tourists waste their time on the road. Below is a detailed analysis of the most relevant literature that forms the foundation of this research:

Machine Learning and Profiling for Tourist Preferences (Recent Studies, 2022-2024): Recent research has shown that machine learning is a powerful tool for understanding what tourists actually want to see. A 2022 study on the "Tourist Trip Design Problem" demonstrated that using Expectation Maximization and clustering algorithms can accurately group tourists based on their categories of interest like beaches or heritage sites [1]. Similarly, research on personalized recommenders (2024) proves that data driven approaches can effectively read past behaviors to suggest the best places without needing a human travel agent [2]. Together, these works provide strong evidence for using machine learning to verify and filter the exact locations a tourist will enjoy.

Algorithms for Itinerary Optimization (Recent Studies, 2017-2022): In terms of planning exactly which order to visit places, advanced algorithms are highly effective. Recent studies have proven that metaheuristic algorithms are the best models for solving the Tourist Trip Design Problem (TTDP) because they handle complex travel times accurately. For example, a 2020 study published in IEEE Access used an Iterated Local Search framework to accurately plan travel itineraries for real world points of interest [3]. Another study (2017) confirmed that using a four step heuristic algorithm, like Genetic Algorithms, significantly improves the quality of personalized day tour routes [4].

Geo-Spatial Boundaries and Multi-Day Planning (2017-2024): The concept of creating travel plans based on a central hotel location has also been explored. Research has demonstrated how advanced optimization models can handle constraints like the transport mode and available time budget [5]. More recent systematic approaches have focused on using algorithms like fuzzy GRASP to solve complex multi day routing problems [6]. Their work confirms that algorithms can be used proactively to create realistic boundaries (radius) and guide tourists without making them travel too far in one day.

In summary, while prior research has addressed individual aspects—such as using machine learning for place recommendation or using Genetic Algorithms for shortest path finding—no single system has combined these into a proactive travel itinerary engine specifically for the Sri Lankan tourism sector. Existing platforms still rely on generic A to B routing and overlook the power of cross validating the tourist's hotel radius with their personal preferences. Our work aims to bridge this gap by designing the Context Aware Spatiotemporal Itinerary Optimization System. This system integrates user preference profiling (Machine Learning) and shortest path optimization to automatically identify the best reachable places and generate day by day travel plans, ensuring a stress free and reliable travel experience in Sri Lanka.

1.2 Research Gap

Existing research on tourist recommendation systems and travel planning has mostly focused on isolated single problems. For example, some studies try to recommend places using user data, while others focus purely on finding the shortest path using distance algorithms. However, there are several critical gaps in developing a system that actively creates a practical and realistic travel plan for independent tourists:

1. Lack of User Profiling and Geo-Spatial Integration: Current systems do not effectively combine what the user likes with where the user is actually staying. There is a lack of systems that integrate Machine Learning (like RNN or clustering) to read user preferences together with dynamic Geo spatial boundaries (radius generation) to limit the travel distance from the hotel.

2. Absence of Time-Constrained, Day-by-Day Routing: Many travel platforms simply display a list of famous places or only show the map from point A to point B. There is a lack of systems that automatically filter reachable places and use Shortest Path algorithms to create a complete, multi day itinerary before the tourist wastes their time on the road.

3. Disconnect Between Recommendation and Physical Limits: Existing recommender models often work by just giving the highest rated places, but they do not check if it is practically possible to visit all of them in one day. There is a gap in using advanced metaheuristic algorithms (like Genetic Algorithms or Iterated Local Search) to balance the user's available travel time, transportation mode, and personal interests at the same time.

According to Table 1, none of the reviewed approaches fully address the combination of ML powered user profiling, dynamic radius generation, shortest path optimization, and multi day planning within a single tourism platform. Therefore, this research addresses these gaps by developing the Context Aware Spatiotemporal Itinerary Optimization System. This system fuses user data with advanced routing algorithms to automatically generate realistic, personalized travel itineraries, ensuring a stress free and reliable travel experience in Sri Lanka.

Table 1: Comparison of the existing research methods and proposed method

Technologies and methods	**[1]**	**[2]**	**[3]**	**[4]**	**[5]**	**[6]**	**[7]**	**Proposed Method (Itinerary Optimization System)**
Point of Interest (POI) Recommendation	✔	✔	✔	✔	✔	✔	✔	✔
User Preference Profiling (Machine Learning)	✔	✘	✔	✘	✘	✘	✘	✔
Shortest Path / Metaheuristic Algorithms	✘	✔	✘	✘	✔	✔	✔	✔
Dynamic Geospatial Boundaries (Radius)	✘	✘	✘	✔	✘	✘	✘	✔
Multi Day / Time Constrained Planning	✘	✔	✘	✘	✘	✘	✔	✔
Multiconstraint Balancing (Preferences+Distance)	✘	✘	✔	✔	✘	✘	✘	✔
Sri Lankan Tourism Context Integration	✘	✘	✘	✘	✘	✘	✘	✔

1.3 Research Problem

Independent tourism is a vital segment of Sri Lanka's economic recovery, with

travelers increasingly exploring the country on their own. However, the current travel planning landscape presents several critical problems that hinder their daily travel experience and practical trip planning:

1. Reliance on Generic Maps and Static Lists: Most existing travel platforms and official tourism websites focus primarily on showing static lists of famous places or simple point A to point B maps. These systems fail to account for the real practical limits of a tourist, such as the maximum distance (radius) they can travel in one day from their booked hotel. As a result, international tourists often plan to visit places that are practically too far away, leading to wasted time on the road and unfinished trips.

2. Disconnect Between User Preferences and Distance Limits: Although tourists have specific interests like beaches, hiking, or historical temples, this information is not connected with distance calculations. While international research has explored recommending places using Machine Learning, these solutions are often designed in isolation. There is no unified system in Sri Lanka that autonomously profiles the user's personal interests and mathematically matches them only with the reachable places inside a generated Geospatial boundary.

3. Poor Scheduling and Lack of Route Optimization: Tourists often have tight, limited days to travel. Trying to visit multiple places without a proper order can ruin an entire day's schedule because they might travel back and forth on the same road unnecessarily. Current platforms do not automatically sort the selected places into a proper, timesaving order. Furthermore, there is a lack of advanced shortest path algorithms (like Genetic Algorithms) in everyday travel apps to provide a clear, optimized travel route, leaving travelers tired and frustrated.

This gap creates an urgent need for a Context Aware Spatiotemporal Itinerary Optimization System that integrates ML powered user profiling, dynamic radius generation, and shortest path routing into a single platform. Such a system can directly save travelers' time on the road, reduce the stress of daily trip planning, and enhance the attractiveness of Sri Lanka as a highly organized and reliable destination for global travelers.

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to design and develop a Context Aware Spatiotemporal Itinerary Optimization System for the Sri Lankan tourism sector, capable of autonomously generating realistic, multi day travel plans for independent tourists. By leveraging Machine Learning for user preference profiling and advanced shortest path algorithms (such as Genetic Algorithms), the system aims to replace generic map applications and static recommendation lists with a dynamic, time constrained travel routing mechanism. Furthermore, the system will utilize Geospatial radius calculations to mathematically limit the travel distance based on the user's hotel location, ensuring they do not travel too far in a single day. Ultimately, this system ensures a practical, stress free, and highly personalized travel experience for international tourists exploring Sri Lanka without wasting their limited time on the road.

2.2 Sub-Objectives

To achieve the main objective, the following specific objectives have been identified:

1. To design and develop a Data Collection and Management Pipeline using the MERN stack (React.js, Node.js, Express, MongoDB) and Python to securely collect and store tourist preferences, hotel locations, and Sri Lankan Point of Interest (POI) data.

2. To implement a Machine Learning User Profiling Module (such as Content Based Filtering or RNN) to perform data analysis, identify the specific travel interests of the tourist (e.g., beaches, heritage sites, or nature), and match them with suitable travel locations without manual filtering.

3. To develop a Dynamic Geospatial Boundary Module, which mathematically calculates a practical travel radius based on the tourist's hotel coordinates, available travel days, and transport mode to make sure they do not plan to travel too far in a single day.

4. To integrate a Shortest Path Optimization Engine using advanced metaheuristic algorithms (such as Genetic Algorithms) into the platform, which arranges the filtered places into a clear, time saving, day by day travel route for the user.

5. To evaluate the accuracy, time efficiency, and effectiveness of the itinerary optimization algorithm and the user profiling logic through controlled testing scenarios involving simulated tourist constraints and real map distances.

3. METHODOLOGY

The methodology for the Context Aware Spatiotemporal Itinerary Optimization System is designed to transform unstructured tourist preferences and map coordinates into actionable, real time travel plans by leveraging Machine Learning, Geospatial calculations, and Shortest Path algorithms.

Data Acquisition and Preprocessing: The process begins with data acquisition and preprocessing, where details about Sri Lankan tourist places (POIs) and user inputs (like hotel location, transport mode, and available days) are collected through the platform. This raw data is securely stored in a MongoDB database. The data is then filtered using basic heuristics to remove closed locations, duplicates, and incomplete map coordinates, ensuring a clean dataset for the routing stages.

Machine Learning User Profiling: The next stage involves the core preference matching module using Python and Machine Learning libraries (such as scikit learn). The preprocessed user data is analyzed to extract critical travel interests: specific categories like beaches, hiking spots, historical temples, or wellness centers. Unlike traditional manual searching, this module interprets the user's past choices or direct inputs to accurately filter and confirm the exact places they will actually enjoy visiting.

Dynamic Radius and Shortest Path Modeling: Once the personalized places are filtered by the ML module, the Routing Engine is activated. This engine first calculates a "Travel Radius" based on the user's hotel coordinates and available time to make sure places are reachable. Then, it utilizes a custom Genetic Algorithm (or similar metaheuristic model). Instead of just showing places on a map, the algorithm tests multiple path combinations to accurately find the shortest distance to connect these places, providing a much more optimized, day by day travel route than static map apps.

Backend Engineering and Optimization: The system's core orchestration is implemented using Node.js and Express.js, which seamlessly handles the communication between the MongoDB database, the Python routing microservice, and the frontend. For performance optimization, frequently

searched Sri Lankan routes and POI details are cached using Redis, ensuring that the system can deliver fast itineraries to multiple tourists simultaneously without server overload. The system logic and API endpoints are rigorously tested using Postman.

Proactive Itinerary Generation Interface: The user facing component is developed using React.js and integrated into the platform as a responsive UI component. When the custom Python Genetic Algorithm finalizes the optimized routing coordinates, the system triggers the itinerary view. To make this mathematical output user friendly, the system sends the final route data to the Gemini API, which acts strictly as an explanation layer. It does not calculate the route. Instead, it automatically generates a clear, human readable travel text (e.g., "Start your Day 1 from Galle Fort, then travel 15 minutes to Unawatuna Beach to save time"). This safely guides travelers without confusing maps.

Evaluation and Performance Testing: Finally, evaluation and performance testing will be conducted in multiple phases. First, the accuracy of the Shortest Path algorithm will be measured against actual map distances and travel times using standard metrics (e.g., total kilometers saved). Second, the system's responsiveness will be evaluated by measuring the processing latency of the Python routing engine and the Gemini API generation cycle. Third, user based testing will be conducted with a group of independent travelers to assess the perceived helpfulness and practicality of the generated travel plans, ensuring a robust and tourist centric travel assistance system.

3.1 Tools and Technologies

Backend Frameworks, AI & Machine Learning Integration: * Node.js with Express: Used for core API handling, system integration, and managing the asynchronous communication between the frontend interface and the AI routing services.

- **Python (ML & Algorithm Core):** Utilized as the primary data science environment to run Machine Learning libraries for user preference profiling and to build the shortest path algorithms (like Genetic Algorithms) for daily route optimization.
- **Gemini API (Google AI):** Acts as the itinerary explanation engine, automatically translating the complex algorithmic map outputs into clear, human readable, day by day travel plans and practical travel suggestions.

Database & Caching: * **MongoDB:** A NoSQL database used for flexible and scalable storage of Sri Lankan tourist places (POIs), map coordinates, and dynamic user travel preferences.

- **Redis:** Employed for real time caching of frequently searched travel routes, generated itineraries, and popular tourist locations to ensure fast, low latency responses for the frontend interface.

API Testing: * **Postman:** Used for testing and validating backend RESTful APIs, ensuring the stability of data pipelines between the MongoDB database, the Python routing algorithm, and the frontend travel plans.

Frontend Interface & UI Components: * **React.js:** Utilized for developing the interactive travel planning UI components that seamlessly integrate into the unified Sri Lankan travel platform.

- **React UI Widgets:** Used to render the proactive itinerary interface and dynamically visualize the day by day travel routes, reachable map boundaries (radius), and personalized tourist places.

Local Storage: * **Browser LocalStorage / State Management:** For local caching of generated travel itineraries, saved tourist preferences, and daily travel plans to improve app performance and provide offline accessibility when tourists have poor internet connections on the road.

Scheduling/Automation:

- **node-cron:** Used for automating periodic system tasks, such as triggering background updates for tourist place details (like open/close times) at regular intervals and clearing expired temporary travel routes from the system.

3.2 System Architecture

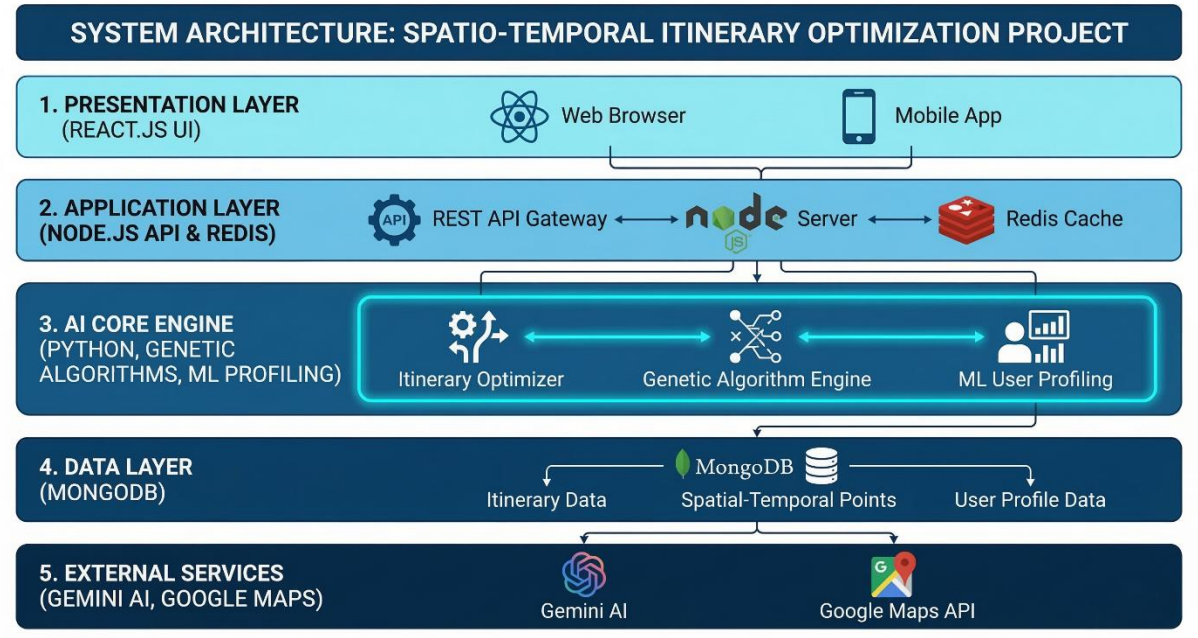


Figure 1 System Architecture

3.3 Component Overview Diagram

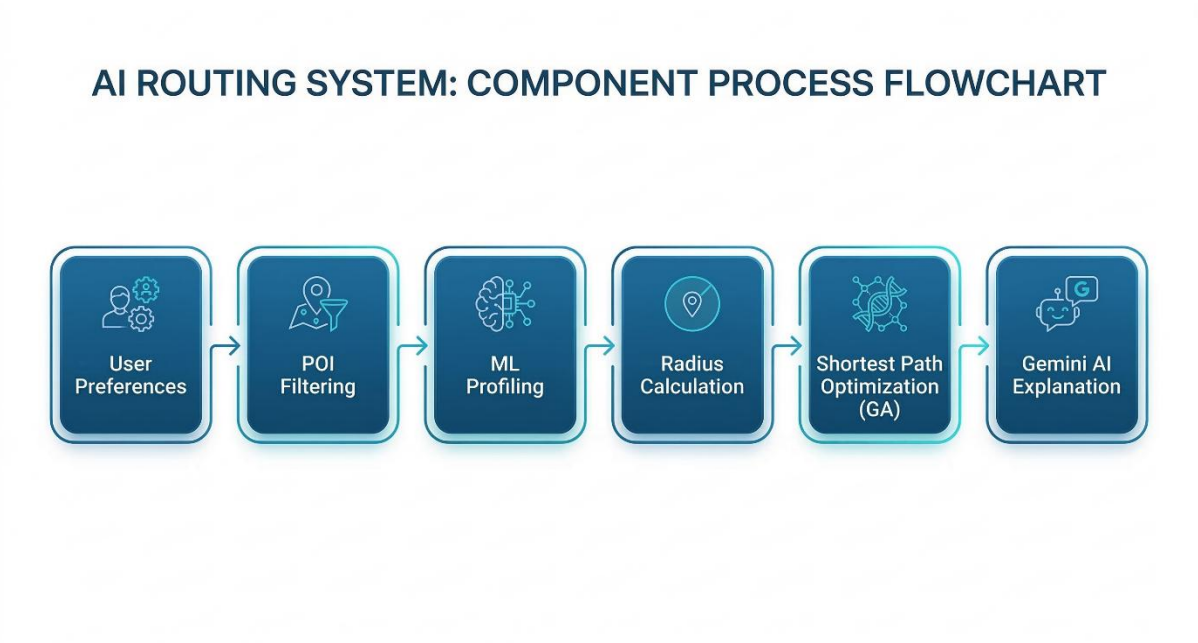


Figure 2 Component Overview Diagram

3.4 Work Breakdown Structure

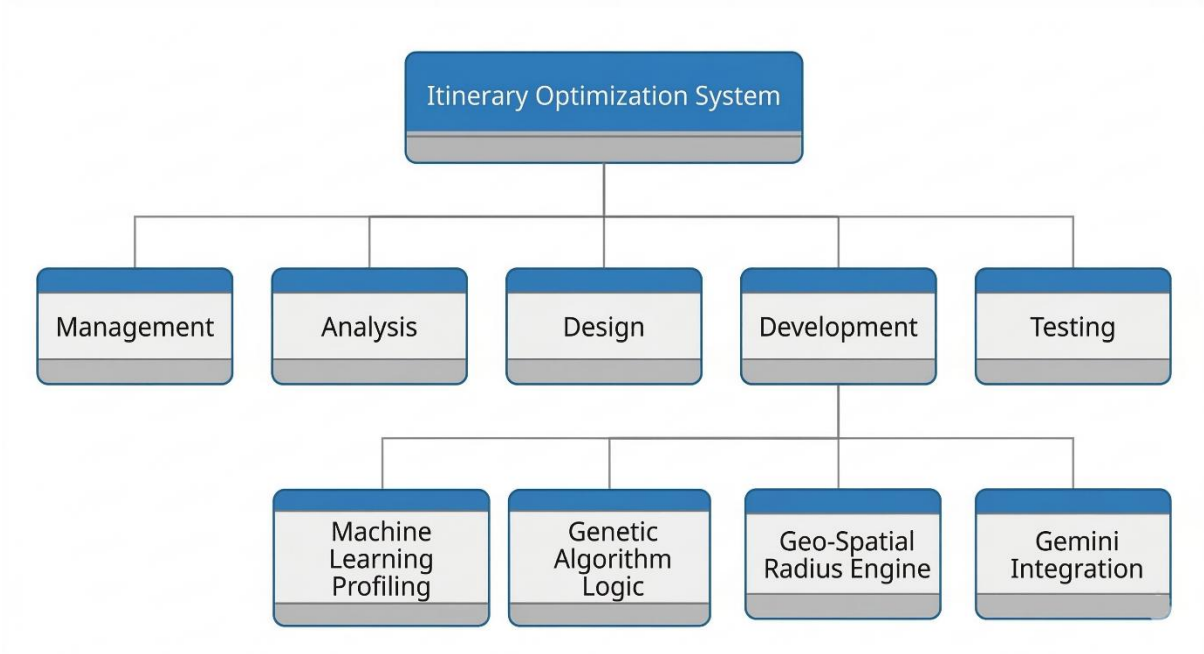


Figure 3Work breakdown structure

3.5 Gantt Chart



Figure 4 Grantt Chart

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

User Preference & Constraint Acquisition: The system should be able to automatically collect and process user inputs, including personal travel interests (e.g., nature, culture), hotel location coordinates, and available travel time constraints.

ML Based Preference Profiling: The system should integrate Python based Machine Learning (such as clustering or content based filtering) to autonomously match the tourist's interests with suitable Sri Lankan Points of Interest (POIs) without manual searching.

Spatiotemporal Routing Engine: The system should utilize advanced shortest path algorithms (such as Genetic Algorithms) to calculate and continuously update optimized travel itineraries based on a dynamically generated travel radius and real world map distances.

Explainable AI (XAI) Itinerary Orchestration: The system should utilize the Gemini API as an XAI layer to translate the complex algorithmic routing outputs into clear, human readable travel plans (e.g., "We suggest visiting Galle Fort first to save 20 mins of travel time") and explain the reasoning behind the route.

Secure Backend Integration: The backend Node.js API should provide secure endpoints for continuous POI data retrieval, user profile management, and seamless communication between the Python routing microservice and the database.

Interactive Itinerary Dashboard: The React frontend interface should dynamically display the day by day travel plans, an interactive map with the travel radius, and personalized location suggestions without disrupting the user's overall trip planning experience.

4.2 Non-Functional Requirements (NFR)

Low Latency & High Performance: The system should process user preference profiling and execute shortest path algorithm calculations with minimal latency, utilizing Redis caching to ensure that the final personalized travel itineraries are delivered to the user in under 3 seconds.

High Scalability: The system architecture must be highly scalable to handle a high volume of concurrent itinerary generation requests and map data processing, especially during peak tourist seasons in Sri Lanka.

Data Privacy & Security: The platform must ensure robust data privacy and security, particularly concerning tourists' hotel locations, personal travel interests, and saved itineraries, through proper authentication (JWT) and secure data encryption.

Usability & Intuitive Design: The user interface for the day by day travel plans and map radius visualization should be intuitive and non intrusive, providing clear, human readable, and actionable travel guidance without overwhelming the user with too much map data.

High Availability: The system should maintain high availability (99.9% uptime) to ensure continuous, uninterrupted trip planning assistance for tourists, even during high traffic periods when many people are searching for travel routes.

Cross-Platform Responsiveness: The itinerary planning interface should be fully responsive and optimized for cross platform mobile environments as an integrated UI component within the broader unified tourism application, as most tourists will use it on their phones.

4.3 System Requirements

Hardware Requirements:

- **Mobile Device:** Minimum 3GB RAM, quad core CPU, active GPS, and stable internet connectivity to receive real time location based itineraries and map updates.
- **Server:** Cloud based VPS with 4+ vCPUs, 8GB+ RAM, and SSD storage to efficiently handle Node.js execution, Python Machine Learning model inferencing (for user profiling), Genetic Algorithm processing (for route optimization), MongoDB operations, and Redis caching.

Software Requirements:

- **Backend Framework:** Node.js with Express.js for core API development and system integration, alongside a Python environment for hosting the Machine Learning profiling and the Genetic Algorithm based routing microservices.
- **Frontend Interface:** React.js for building the dynamic Itinerary Planning interface, seamlessly integrated into the unified mobile platform as responsive UI components for the travel dashboard.
- **Database:** MongoDB (NoSQL) for flexible and scalable storage of Sri Lankan Points of Interest (POIs), map coordinates, user travel preferences, and historical itinerary logs.
- **Caching Engine:** Redis for real time caching of generated travel routes, frequently visited locations, and distance calculation results to ensure ultra low latency.
- **Machine Learning & Optimization Core:** Python with scikit learn for user preference profiling and custom implementations of shortest path metaheuristic algorithms (e.g., Genetic Algorithms) for spatiotemporal itinerary optimization.
- **Explainable AI (XAI) Engine:** Gemini API (Google AI) for acting as the Explainable AI layer, automatically translating the complex algorithmic routing data into clear, human readable travel plans and personalized suggestions.
- **API Testing:** Postman for testing, documenting, and validating backend RESTful APIs and the integration between the routing microservice and the frontend.
- **Task Scheduling:** node cron for automating periodic system tasks, such as updating the database with new tourism locations or refreshing seasonal opening/closing times for Sri Lankan POIs.

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